ARGOS H1 engine ECU v2.5 (Teensy 4.1, separate from the flight computer, CAN 1 Mbit/s)

Eight hierarchical sheets (v2.3 adds Pintle_Rail and Sensors_B); nets connect by global labels and power symbols.

Generated 2026-10-13 by hardware/kicad_gen/make_ecu_kicad.py from hardware/kicad_gen/ecu_design.py, then editable by hand.

KiCad used: 7.0.11 (Ubuntu 24.04 package 7.0.11+dfsg-1build4). Files use the KiCad 7 format and open in KiCad 7, 8 and 9 (8/9 upgrade on save).

ERC: KiCad 7 eeschema ERC run headless (kicad-cli 7 has no sch erc); report: erc/argos_ecu_erc.rpt.

All symbols live in the project library argos_ecu.kicad_sym (flattened copies of KiCad 7.0.11 stock symbols + custom).

Status: design schematic only. Not built, no PCB layout routed, no bench test. ERC clean is not hardware validation.

Safety chain: arming plug (physical) -> fused ARM bus switch = ARM_REQ AND watchdog OK AND OVP_OK -> ARM-only 5 V gate rail.

OVP_OK: LMV331 comparator on PC_SIG (31.47 bar g on the 070BG, ratiometric) clears a 74LVC1G175 latch; only an ARM_REQ rising edge sets it.

v2.3: MV holding magnet gate = MV latch command AND PRAIL_OK (LTC4365 OV disconnect of the pintle rail, U306).

v2.3: GSE heater interlock J616 closes only with the firmware permit AND no arming plug AND ARM bus off AND watchdog OK.

Placeholder footprints (not for fabrication): Teensy socket outline (check), Pololu wire pads, TPS552882 RPM0026A pads.

Power

File: power.kicad_sch

MCU_CAN

File: mcu_can.kicad_sch

Safety_Arming

File: safety_arming.kicad_sch

Actuators

File: actuators.kicad_sch

Igniter_Valves

File: igniter_valves.kicad_sch

Sensors

File: sensors.kicad_sch

Pintle_Rail

File: pintle_rail.kicad_sch

Sensors_B

File: sensors_b.kicad_sch

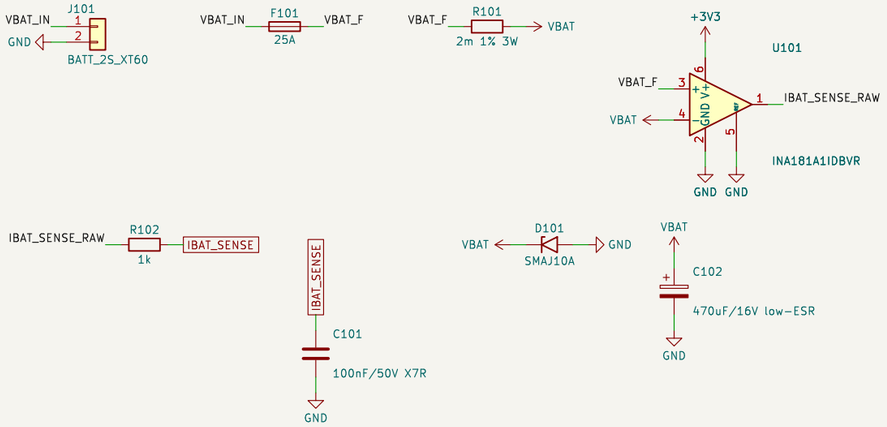
Status: design schematic, not hardware-validated
KiCad used: 7.0.11 (Ubuntu 24.04 package 7.0.11+dfsg-1build4)
Root: overview and sheet index
ARGOS H1 engine - 06-electronics

Sheet: /
File: argos_ecu.kicad_sch

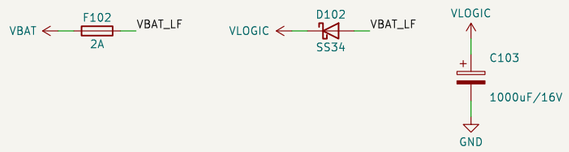
Title: ARGOS engine ECU v2.5 - root

Size: A3	Date: 2026-10-13	**Rev: v2.5**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/9

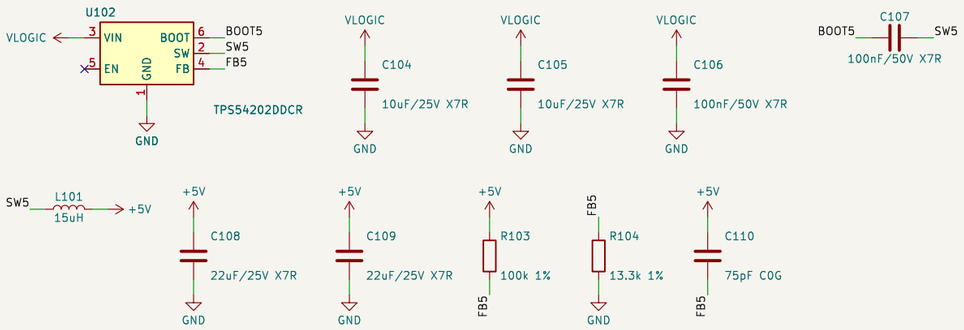
Battery input, main fuse, current sense, transient clamp



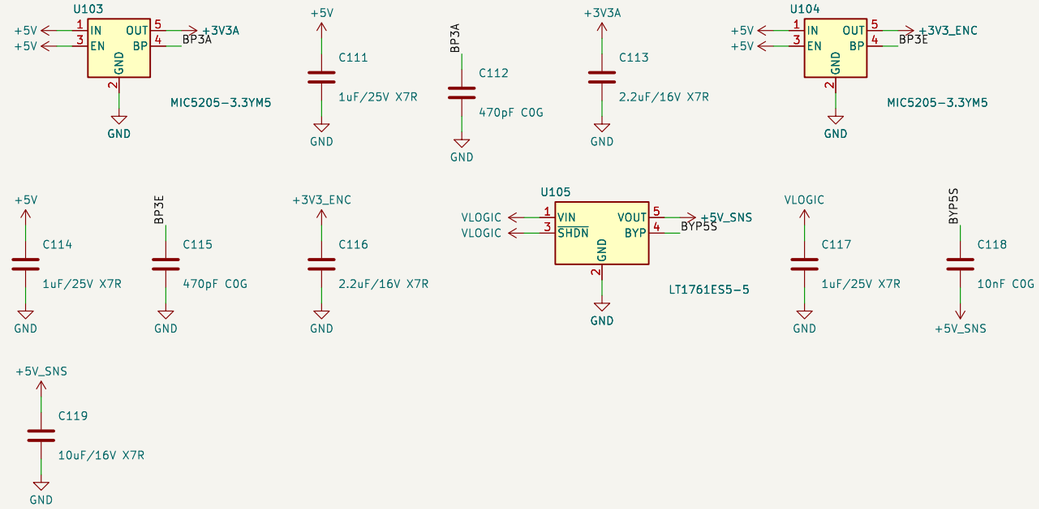
Logic supply path: 2 A fuse, Schottky, 1 mF hold-up



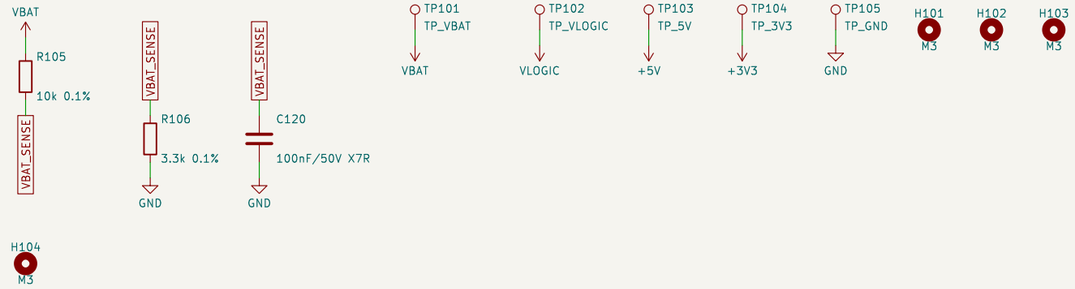
5 V buck TPS54202 (TI SLVSD26C 5 V design: 15 uH, 2x22 uF, 100k/13.3k, 75 pF)

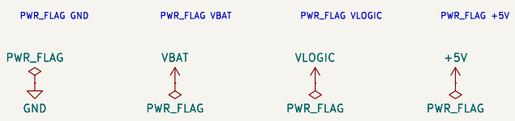


Low-noise LDOs: +3V3A (ADC AVDD), +3V3_ENC (encoder cables), +5V_SNS (transducers)



Battery voltage sense, test points, mounting





Status: design schematic. ERC clean is NOT hardware validation. Not built, not tested.
KiCad used: 7.0.11 (Ubuntu 24.04 package 7.0.11+dfsg-1build4)
Generated by hardware/kicad_gen/make_ecu_kicad.py (2026-10-13); edit freely in KiCad

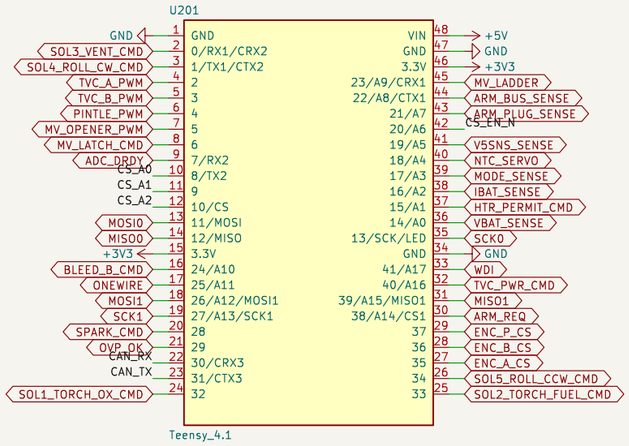
ARGOS H1 engine — 06-electronics

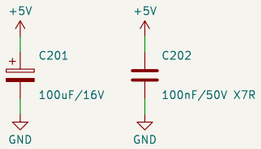
Sheet: /Power/
File: power.kicad_sch

Title: ARGOS ECU v2.5 — Power input, protection, rails

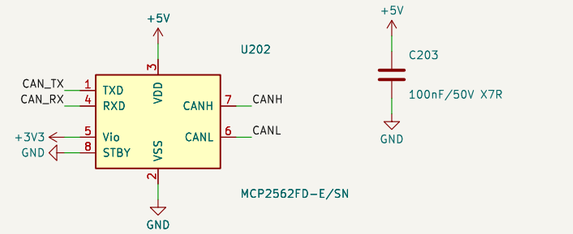
Size: A3	Date: 2026-10-13	**Rev: v2.5**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 2/9

MCU: PJRC Teensy 4.1 on 2x24 sockets (cut VUSB–VIN pads; USB is service only)

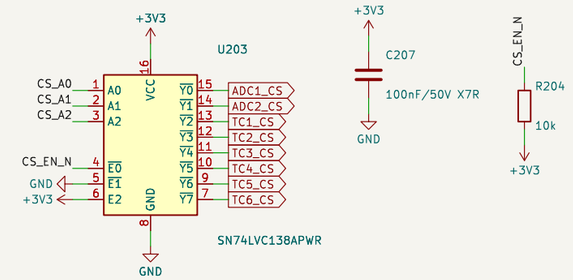


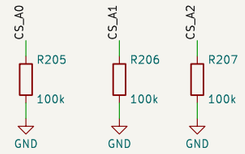


CAN FD transceiver MCP2562FD (CAN3, 1 Mbit/s classic, FD-ready), split termination, ESD



SPI0 chip-select decoder SN74LVC138A (v2.3): CS_A2..A0 + CS_EN_N -> ADC1, ADC2, TC1–TC6; all CS high in reset





Status: design schematic. ERC clean is NOT hardware validation. Not built, not tested.
KiCad used: 7.0.11 (Ubuntu 24.04 package 7.0.11+dfsg-1build4)
Generated by hardware/kicad_gen/make_ecu_kicad.py (2026-10-13); edit freely in KiCad

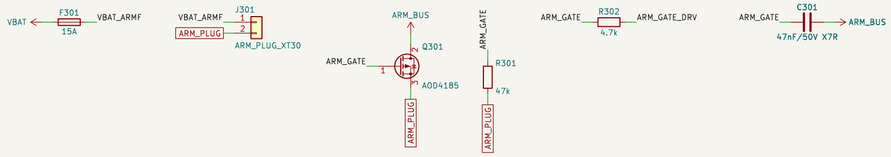
ARGOS H1 engine — 06-electronics

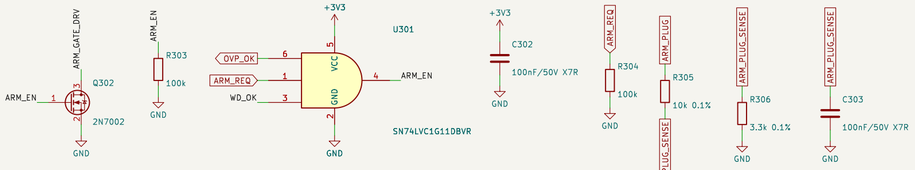
Sheet: /MCU_CAN/
File: mcu_can.kicad_sch

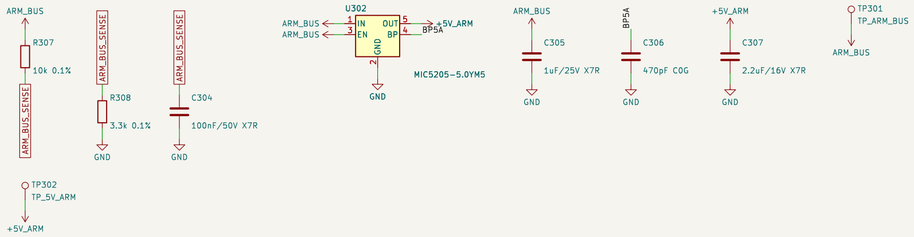
Title: ARGOS ECU v2.5 — MCU Teensy 4.1 and CAN

Size: A3	Date: 2026-10-13	**Rev: v2.5**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 3/9

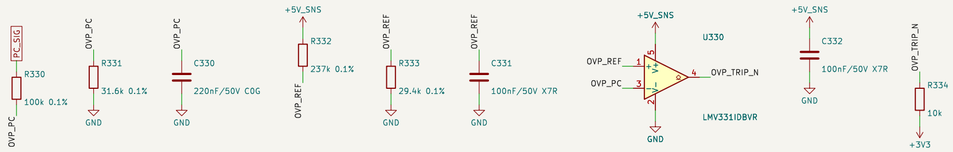
Arming plug (physical break) and ARM high-side switch (soft start)

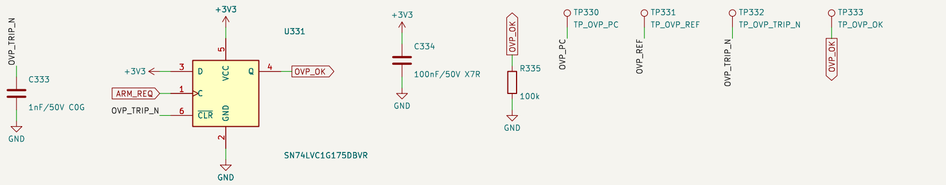




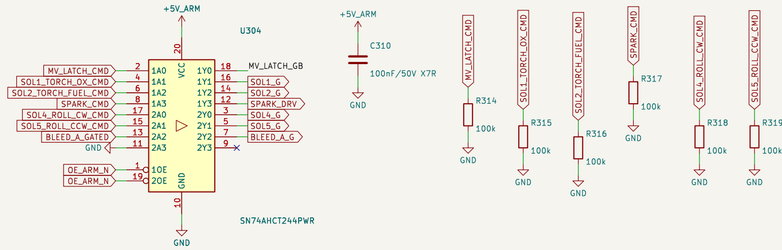


Hardware chamber over-pressure trip (~31 bar g): LMV331 -> SN74LVC1G175 latch -> ARM AND; re-armed only by an ARM_REQ rising edge

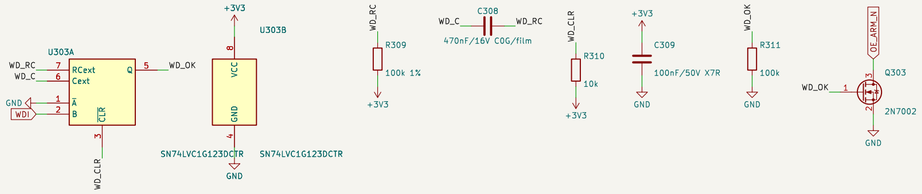


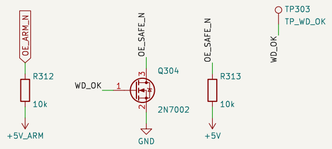


ARM-gated gate buffer U304 SN74AHCT244 on +5V_ARM (/OE = watchdog)

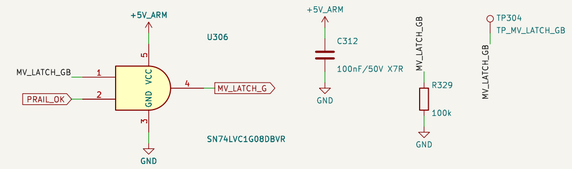


Independent hardware watchdog SN74LVC1G123 (tw = K R C = 1.0–1.1 x 100k x 470nF = 47–52 ms)

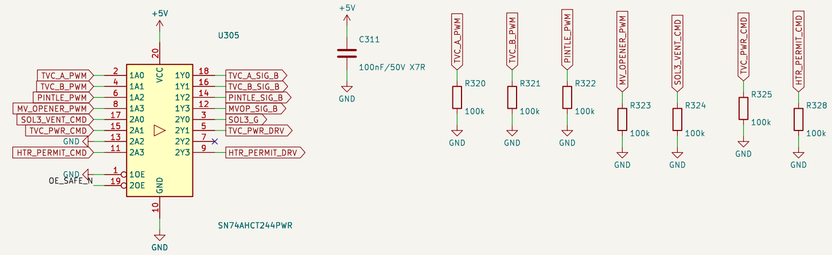




MV holding-magnet permit (v2.3): MV_LATCH_G = MV latch command AND PRAIL_OK (LTC4365 /FAULT, pintle sheet)



SAFE-allowed buffer U305 SN74AHCT244 on +5V: servo PWM 3.3->5 V (always on), vent + TVC power (watchdog)





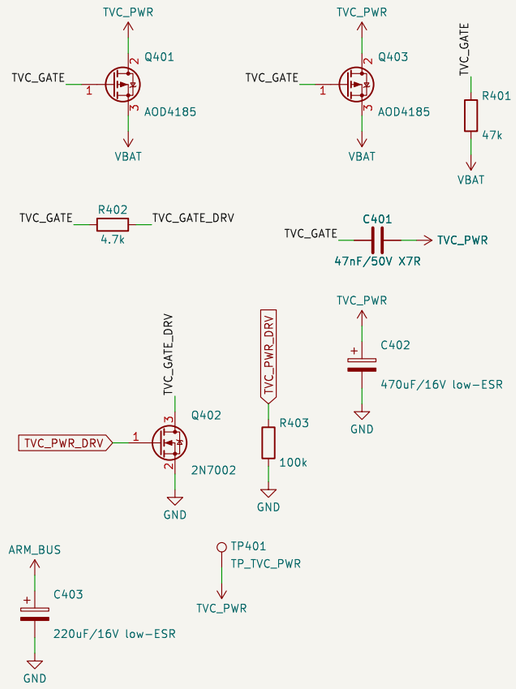
Status: design schematic. ERC clean is NOT hardware validation. Not built, not tested.
KiCad used: 7.0.11 (Ubuntu 24.04 package 7.0.11+dfsg-1build4)
Generated by hardware/kicad_gen/make_ecu_kicad.py (2026-10-13): edit freely in KiCad
ARGOS H1 engine — 06-electronics

Sheet: /Safety_Arming/
File: safety_arming.kicad_sch

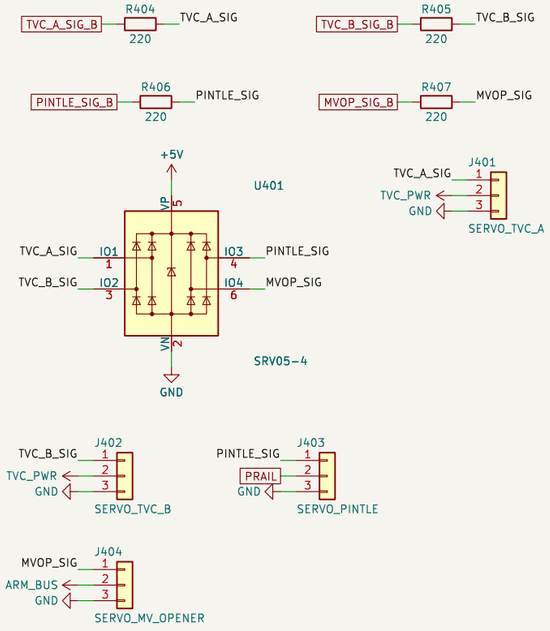
Title: ARGOS ECU v2.5 — Arming, watchdog, output buffers

Size: A2	Date: 2026-10-13	**Rev: v2.5**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 4/9

TVC servo power switch (watchdog-gated, soft start): 2 x AOD4185 in parallel, no branch fuse (v2.3), reservoirs



Servo connectors (Micro-Fit 3.0: 1 signal, 2 V+, 3 GND), 220 ohm series, ESD





Status: design schematic. ERC clean is NOT hardware validation. Not built, not tested.
KiCad used: 7.0.11 (Ubuntu 24.04 package 7.0.11+dfsg-1build4)
Generated by hardware/kicad_gen/make_ecu_kicad.py (2026-10-13); edit freely in KiCad
ARGOS H1 engine — 06-electronics

Sheet: /Actuators/
File: actuators.kicad_sch

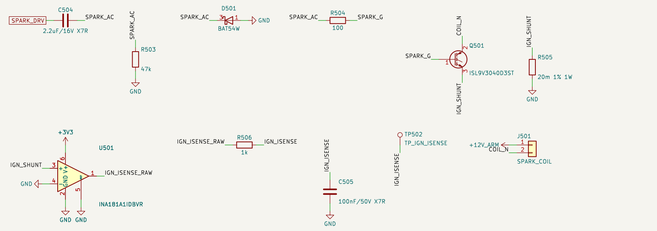
Title: ARGOS ECU v2.5 — Servo actuators, TVC power switch

Size: A3	Date: 2026-10-13	**Rev: v2.5**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 5/9

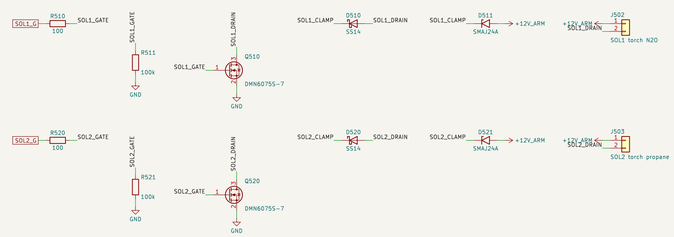
12 V boost on ARM bus (Pololu U3V70F12): spark coil, torch and roll solenoids



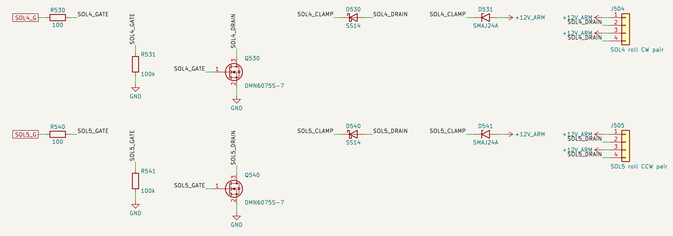
Spark coil driver: logic-level ignition IGBT, AC-coupled gate (hardware max dwell), current sense



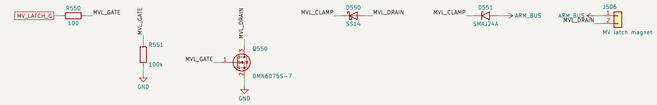
Torch solenoids (12 V, ARM-gated)



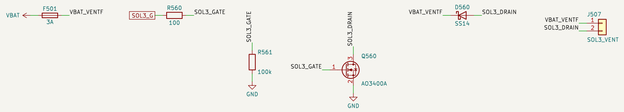
Roll RCS provision (LEAD 2026-10-12: 2 reserved channels, one N2O-vapour thruster pair each; 12 V, ARM-gated; <= 0.75 A per coil ASSUMED, no valve chosen)



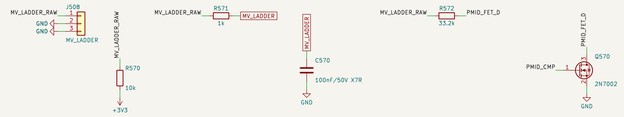
Main-valve holding magnet (ARM bus, fast release; gate from U306 = command AND PRAIL_OK since v2.3)



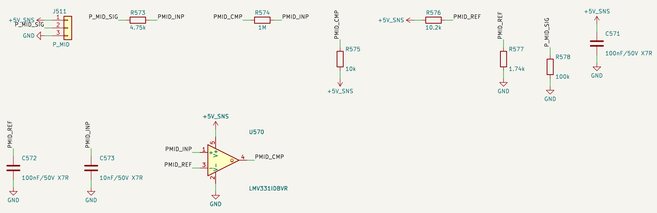
Tank vent solenoid (VBAT, PWM-capable, SAFE-allowed, watchdog-gated)



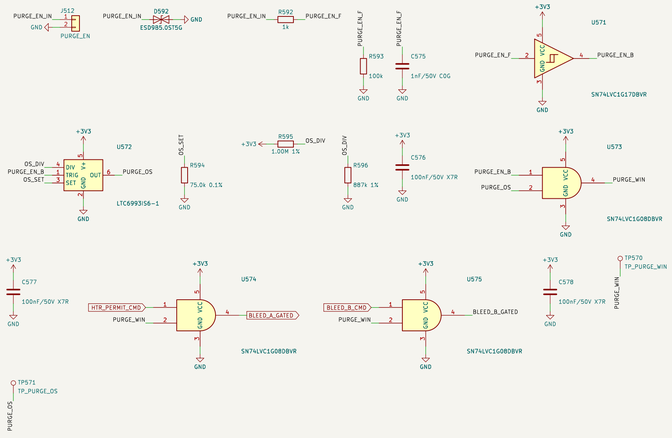
v2.4 MV switch + P_mid ladder on pin 23 (R0/R1/R2 in the J508 mating backshell; calc/hbleed_provision_v1.py)



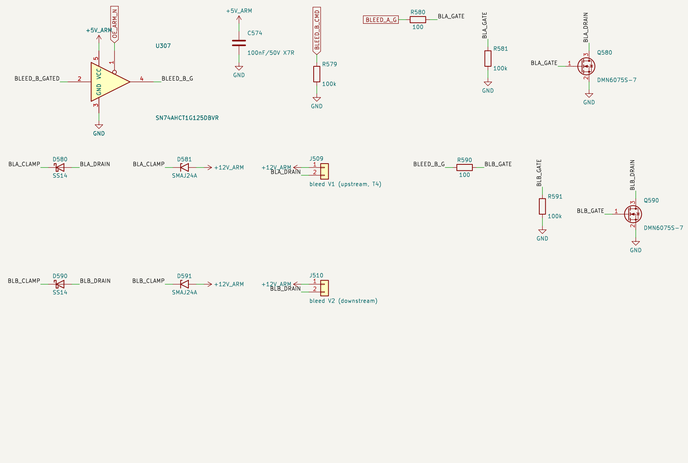
v2.4 P_mid comparator (between bleed V1 and V2): 170BG on J511 -> LMV331, rise 9.86 / fall 8.86 bar g



v2.5 hardware purge window (LEAD CHANGELOG 166 (3)): PURGE_WIN = FC PURGE_EN AND one-shot 3.05–3.24 s; V1 = pin 15 AND PURGE_WIN (U574), V2 = pin 24 AND PURGE_WIN (U575): window chain on +3V3

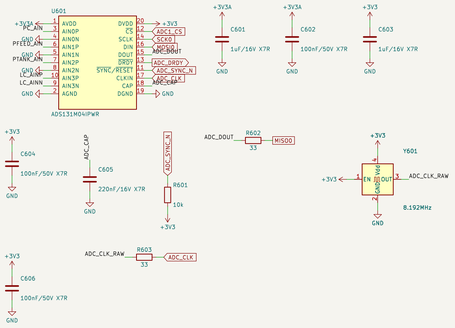


head-end bleed valves V1 / V2 (NC, in series, fixed side: +12V_ARM low-side: V2 buffer U307 separate from U304; v2.5 inputs gated by U574/U575)

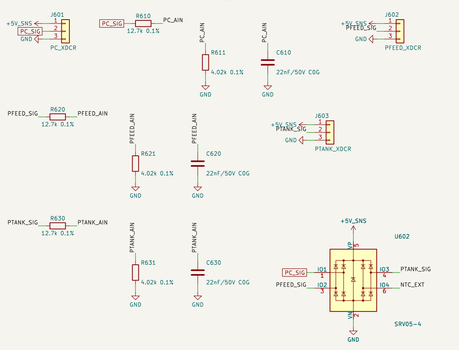


Status: design schematic. ERC clean is NOT hardware validation. Not built, not tested.
KiCad used: 7.0.11 (Ubuntu 24.04 package 7.0.11+dfsg-1build4)
Generated by hardware/kicad_gen/make_ecu_kicad.py (2026-10-13); edit freely in KiCad
ARGOS M1 engine — 06-electronics
Sheet: /Igniter_Valves/
File: Igniter_Valves.kicad_sch
Title: ARGOS ECU v2.5 — 12 V boost, spark, solenoid drivers
Size: A1 | Date: 2026-10-13 | **Rev: v2.5**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4 | Id: 6/9

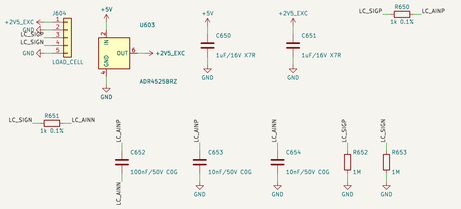
24-bit ADC1 ADS131M04 (4 kSPS, decimated to 1 kHz in firmware), 8.192 MHz clock shared with ADC2



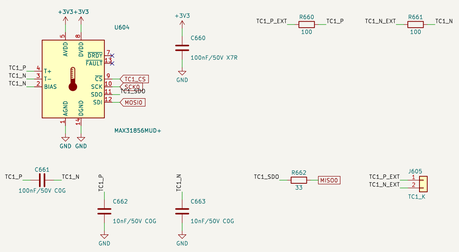
Pressure transducers (0.5–4.5 V ratiometric on +5V_SNS): divider + anti-alias RC



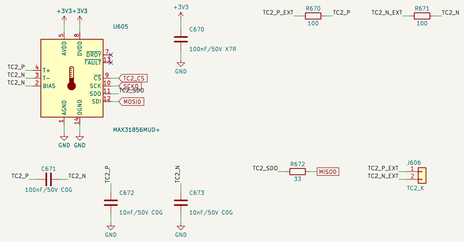
Load cell (test stand, ADC CH3 PGA 64–128) with 2.5 V precision excitation



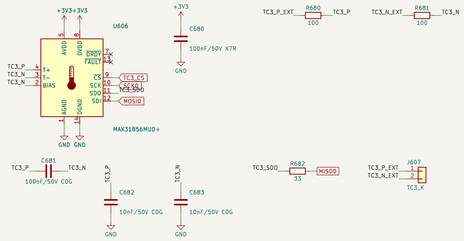
Thermocouple 1 (case forward): MAX31856, K type, 100 ohm + 100 nF + 10 nF filter (Maxim 19-7534 Fig. 8)



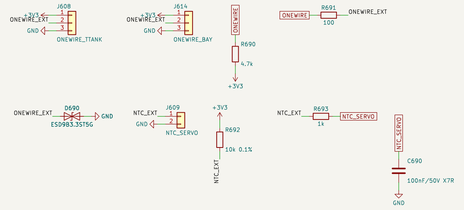
Thermocouple 2 (case aft): MAX31856, K type, 100 ohm + 100 nF + 10 nF filter (Maxim 19-7534 Fig. 8)



Thermocouple 3 (nozzle holder): MAX31856, K type, 100 ohm + 100 nF + 10 nF filter (Maxim 19-7534 Fig. 8)



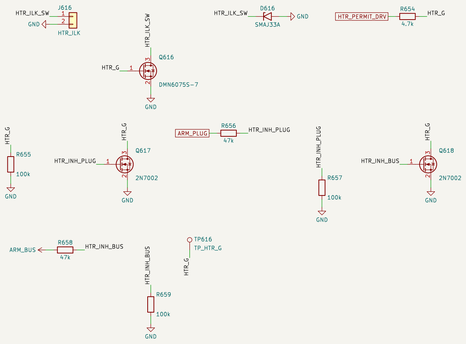
1-Wire DS18B20 bus (tank wall #1/#2, bay #3) and pintle-servo bracket NTC



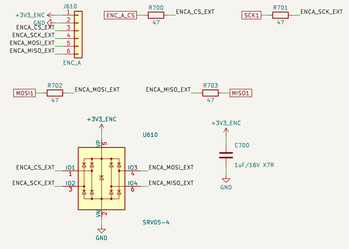
Mode strap J620 (v2.3: firmware v2.8): resistor code on Teensy A3 (pin 17), ratiometric to +3V3



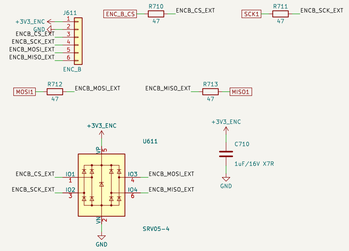
GSE tank-heater interlock J616 HTR_ILK (v2.3, LEAD 2026-10-12 condition (b)): coil loop closed only with the firmware permit (U305, watchdog OK) AND no arming plug AND ARM bus off



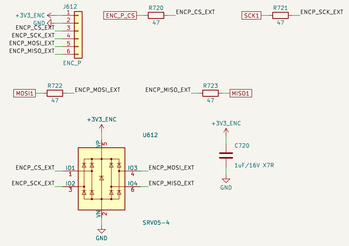
Encoder gimbal axis A: AS5048A (3.3 V mode) over shielded cable, 47 ohm series, ESD



Encoder gimbal axis B: AS5048A (3.3 V mode) over shielded cable, 47 ohm series, ESD



Encoder pintle crank: AS5048A (3.3 V mode) over shielded cable, 47 ohm series, ESD



Status: design schematic. ERC clean is NOT hardware validation. Not built, not tested.
KiCad used: 7.0.11 (Ubuntu 24.04 package 7.0.11+dfsg-1build4)
Generated by hardware/kicad_gen/make_ecu_kicad.py (2026-10-13); edit freely in KiCad

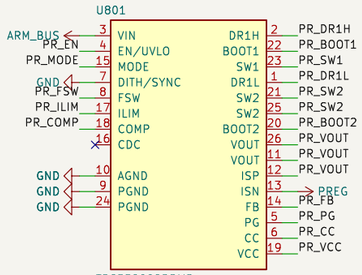
ARGOS M1 engine — 06-electronics

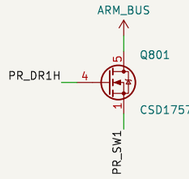
Sheet: /Sensors/
File: sensors.kicad_sch

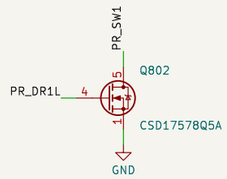
Title: ARGOS ECU v2.5 — Sensors A, encoders, strap, heater ILK

Size: A1	Date: 2026-10-13	**Rev: v2.5**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 7/9

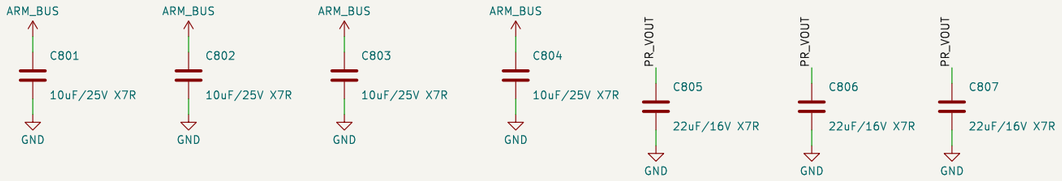
Pintle servo rail 7.49 V: TPS552882 four-switch buck-boost (boost leg integrated, buck leg = external Q801/Q802), FPWM 400 kHz, output current limit 4.0-4.33 A on R807

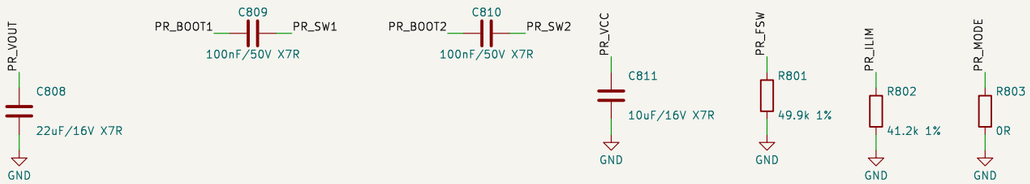


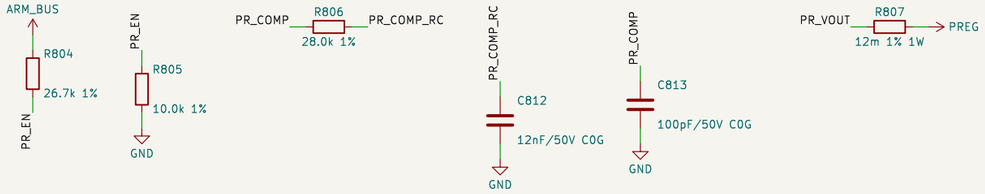


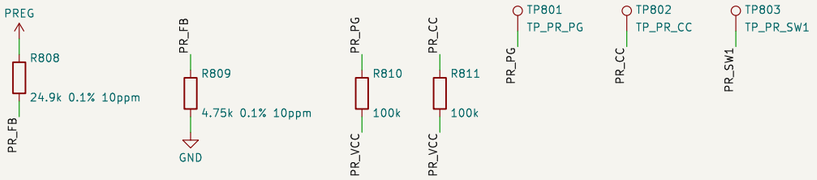








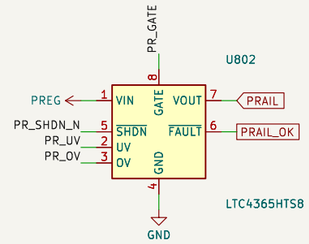


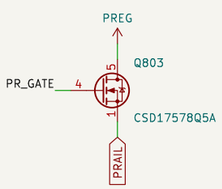


PWR_FLAG PREG

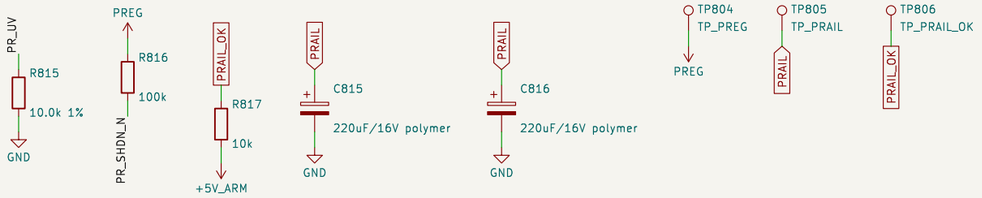


OV/UV disconnect LTC4365 (LEAD 2026-10-13): OV 8.20 V (8.06-8.34 V), filter tau 9.4 us; /FAULT -> PRAIL_OK -> U306 (MV latch released in hardware); servo-side rail PRAIL to J403 and ADC2 CH1

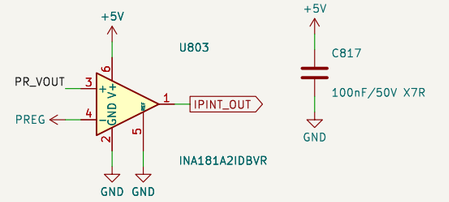








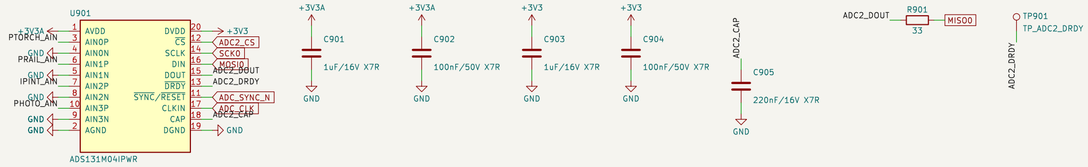
Pintle rail current sense (firmware v2.6 stall current path): INA181A2 x50 across R807



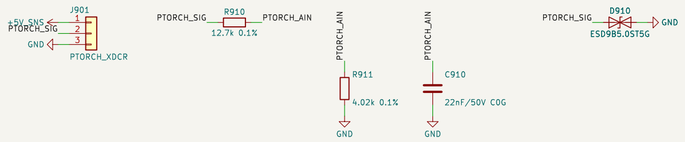
Status: design schematic. ERC clean is NOT hardware validation. Not built, not tested.
KiCad used: 7.0.11 (Ubuntu 24.04 package 7.0.11+dfsg-1build4)
Generated by hardware/kicad_gen/make_ecu_kicad.py (2026-10-13); edit freely in KiCad
ARGOS H1 engine - 06-electronics

Sheet: /Pintle_Rail/		
File: pintle_rail.kicad_sch		
Title: ARGOS ECU v2.5 - Pintle rail TPS552882 + LTC4365		
Size: A3	Date: 2026-10-13	**Rev: v2.5**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 8/9

ADC2 ADS131M04 (v2.3): CH0 P_torch, CH1 PRAIL, CH2 pintle current, CH3 FA-1 photodiode head (v2.5; PREG retired to TP902); clock, SYNC and SPI0 shared with ADC1; OSR 4096 -> 1 kSPS, read once per 1 ms tick



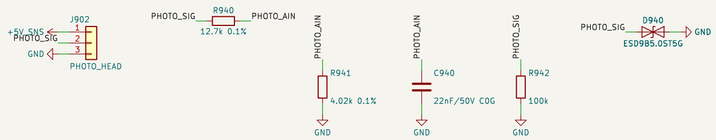
Torch chamber pressure P_torch (RC-01 flame gate, firmware v2.7): TE M3200 070BG on a standoff, Pc chain



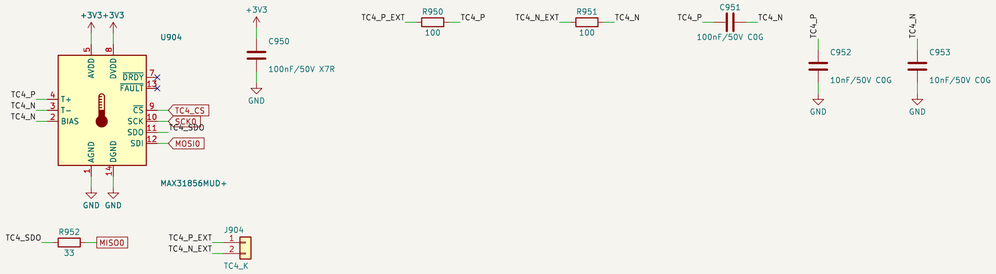
Pintle rail sense: PRAIL (servo side, firmware loss detection < 5.5 V), rail current; PREG test point only (v2.5)



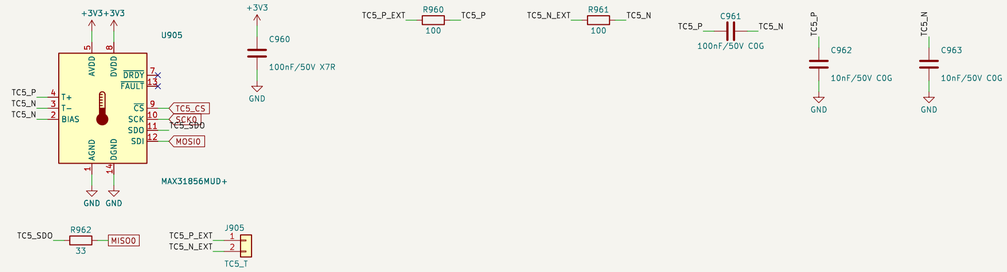
v2.5 FA-1 flame-light head (photodiode + log transimpedance amplifier, remote head on J902) -> ADC2 CH3 (firmware v2.10 inputs::photo_V, valid 0.02-4.80 V at the head)



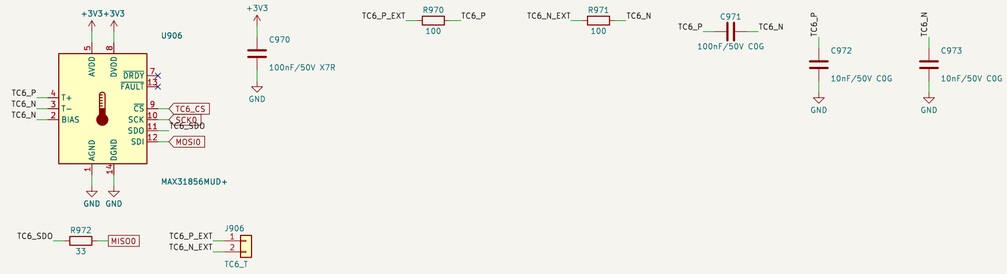
Thermocouple 4 (bore-top / manifold (firmware v2.7, CF-3/T1 evidence)): MAX31856, type K, 100 ohm + 100 nF + 10 nF filter (Maxim 19-7534 Fig. 8)



Thermocouple 5 (MSS-1 inlet-block manifold state sensor (LEAD 2026-10-13)): MAX31856, type T, 100 ohm + 100 nF + 10 nF filter (Maxim 19-7534 Fig. 8)



Thermocouple 6 (MSS-2 NC-valve-outlet manifold state sensor (LEAD 2026-10-13)): MAX31856, type T, 100 ohm + 100 nF + 10 nF filter (Maxim 19-7534 Fig. 8)



Status: design schematic. ERC clean is NOT hardware validation. Not built, not tested.
KiCad used: 7.0.11 (Ubuntu 24.04 package 7.0.11+dfsg-1build4)
Generated by hardware/kicad_gen/make_ecu_kicad.py (2026-10-13): edit freely in KiCad
ARGOS H1 engine — 06-electronics

Sheet: /Sensors_B/
File: sensors_b.kicad_sch

Title: ARGOS ECU v2.5 — ADC2, P_torch, rail sense, TC4-TC6

Size: A2	Date: 2026-10-13	**Rev: v2.5**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 9/9